

FEASIBLE SEARCH BEHAVIOR

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INTRODUCTION

PANDORA'S PROBLEM

- ▶ An agent sequentially searches for the best alternative. (e.g., shopping and job search)
- ▶ Weitzman 's (1979) search model (**Pandora's problem**) captures this scenario.
 - ▶ Multiple **asymmetric boxes** (options) with unknown **rewards**.
 - ▶ A decision maker, **Pandora**, engages in sequential search.
 - ▶ Inspecting (opening) a box is **costly**, but it **fully** reveals the reward in it.

RESEARCH QUESTION

- ▶ In applications, agents may not fully observe the reward by inspections.
 - ▶ Rather, they would observe **partial information** on it.
 - ▶ The underlying **information structures** would affect her search behavior.

Research Question

What search behavior can be rationalized with some information structure?

- ▶ We add incomplete information to **Weitzman 's (1979)** model.
- ▶ We explore an interaction between **information** and **search behavior**.

RESULTS

- ▶ Search behavior = Probabilities that each box is opened.
- ▶ Theorem 1 derives **all feasible search behaviors** arises under **some** info. structures.
 - ▶ The feasible set forms a polytope.
 - ▶ A **single information structure** rationalizes any feasible search behavior.
 - ▶ This **minimizes** Pandora's welfare among all information structures.
- ▶ Theorem 2 characterizes all pairs of search and **choice behaviors**. (same implications)
- ▶ Theorem 3 considers **price competition** among sellers of options. (same feasible set)

IMPLICATIONS

- ▶ Information-free prediction:
 - ▶ We can derive the **prediction set** of behavior without knowing information structures.
- ▶ Information design:
 - ▶ For any objectives over search (and choice) behaviors, **optimal info. structure** is derived.
 - ▶ It minimizes Pandora's welfare **as if the designer provides no information**.
- ▶ Partial identification:
 - ▶ Econometrician wishes to identify the primitives from observed data of search behavior.
 - ▶ We can derive the **identified set** of primitives under **minimal informational assumptions**.

LITERATURE: ROBUST PREDICTIONS/INFORMATION DESIGN

- ▶ Bayes Correlated Equilibrium:
 - ▶ Pioneered works: [Bergemann and Morris \(2013\)](#) and [Bergemann and Morris \(2016\)](#).
 - ▶ Extensive form games: [Makris and Renou \(2023\)](#) and [Doval and Ely \(2020\)](#).
- ▶ Focusing on action marginal and implications to partial identification:
 - ▶ [Doval *et al.* \(2024\)](#) and [De Oliveira and Lamba \(2025\)](#).
- ▶ Ours: Same analysis but focusing on search models.
 - ▶ Search models are complicated as information depends on past actions.
 - ▶ We derive the strongest reduction principle.

LITERATURE: INFORMATION AND SEARCH

- ▶ The role of information in **random search**:
 - ▶ Information design: [Dogan and Hu \(2022\)](#) and ?.
 - ▶ Competition: [Board and Lu \(2018\)](#), [Lyu \(2023\)](#), [Mekonnen and Pakzad-Hurson \(2025\)](#).
- ▶ The role of information in **ordered search**:
 - ▶ Competition: [Au and Whitmeyer \(2023\)](#) and [Au and Whitmeyer \(2024\)](#).
- ▶ Ours: Feasible search (and choice) behavior (with pricing) under general asymmetry.

LITERATURE: EMPIRICAL WORKS

- ▶ Identification and estimation of search costs and preferences:
 - ▶ Jolivet and Turon (2019) and Moraga-González *et al.* (2023) for example.
 - ▶ Survey: Ursu *et al.* (2025).
- ▶ Recent wave is the partial identification under weak informational assumptions: e.g.,
 - ▶ Export markets: (Dickstein and Morales, 2018)
 - ▶ Entry games: (Magnolfi and Roncoroni, 2023)
 - ▶ Prescription drug choices (Dickstein *et al.*, 2024)
 - ▶ Spatial voting (Gualdani and Sinha, 2024)
 - ▶ Migration (Porcher *et al.*, 2024)
- ▶ Ours: The identified set with weak informational assumptions in search model.

MODEL

WEITZMAN (1979)

- ▶ Risk-neutral decision maker (**Pandora**) and finite **boxes** $i = 1, 2, \dots, n$.
 - ▶ Each box i contains a random **reward** $x_i \in \{\underline{x}_i, \bar{x}_i\}$ with $Prob(x_i = \bar{x}_i) = \lambda_i$ and mean μ_i .
 - ▶ It costs $c_i \in (0, \lambda_i(\bar{x}_i - \underline{x}_i))$ to **open** box i and observe some **signal** s_i about its reward x_i .


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- ▶ Pandora's strategy determines **whether or not to open which box** at each stage.
 - ▶ She can choose one of **any opened boxes** when stopping search.
- ▶ A **search behavior** $v \in \mathbb{R}^n$ describes each probability v_i that box i is opened.

INFORMATION STRUCTURE

- ▶ A **signal** s_i is a random variable whose realization depends on x_i .
 - ▶ Upon observing s_i , Pandora forms expectation $y_i = \mathbb{E}[x_i | s_i]$.
- ▶ In our model, posterior mean $y_i = \mathbb{E}[x_i | s_i]$ is the only payoff-relevant information.
 - ▶ We model an **information structure** (of box i) as a CDF $F_i \in \Delta([\underline{x}_i, \bar{x}_i])$.
 - ▶ Some signal induces F_i if and only if F_i has mean μ_i (**Blackwell, 1953**).
 - ▶ Call such F_i **feasible** and let $F = (F_i)_i$.

PANDORA'S OPTIMAL STRATEGY

- ▶ Once fix CDFs $F_1, \dots, F_n$, our model is mathematically identical to [Weitzman \(1979\)](#).
- ▶ A solution z_i to the following equation is called **reservation value** of box i :

$$\underbrace{c_i}_{\text{Search cost}} = \underbrace{\int_{z_i}^{\bar{x}_i} (y_i - z_i) dF_i(y_i)}_{\text{Gain from search}}$$

Pandora's Rule ([Weitzman \(1979\)](#))

Search boxes in descending order of reservation values
until a posterior exceeding the reservation value of any unopened box.

FEASIBLE SEARCH BEHAVIOR

Definition

A search behavior v is feasible if there exists a feasible CDF profile F that induces v .

- ▶ We are interested in the set of **all feasible search behaviors** and its properties.
- ▶ Formula: Suppose that $z_1 > z_2 > \dots > z_n$ under some CDFs $F_1, \dots, F_n$. Then,

$$F_1(z_i-) \cdot F_2(z_i-) \dots F_{i-1}(z_i-) \leq v_i \leq F_1(z_i+) \cdot F_2(z_i+) \dots F_{i-1}(z_i+),$$

where $F_j(z_i-)$ is the left limit of F_j and $F_j(z_i+)$ is the right limit of F_j .

ASSUMPTION

- Once can derive the range of **feasible reservation values**:

$$\underline{z}_i = \mu_i - c_i \leq z_i \leq \bar{x}_i - \frac{c_i}{\lambda_i} = \bar{z}_i.$$

Assumption

$$\bar{z} \geq \underline{z} \text{ where } \underline{z} = \max_i \underline{z}_i \text{ and } \bar{z} = \min_i \bar{z}_i.$$

- Under this assumption, **any search order can be induced** by some CDFs.
- This is satisfied if search costs are small or asymmetry across boxes is weak.

RESULTS

NOTATIONS

- ▶ An order $\pi : \underbrace{\{1, \dots, n\}}_{\text{order}} \rightarrow \underbrace{\{1, \dots, n\}}_{\text{box}}$ is a permutation over the set of boxes.
- ▶ Define, for each i ,

$$\bar{v}_{\pi,i} = \prod_{j < i} p_{\pi(j)}, \quad p_i = 1 - \frac{c_i}{\underline{x}_i - \underline{z}},$$
$$\underline{v}_{\pi,i} = \prod_{j < i} q_{\pi(j)}, \quad q_i = 1 - \frac{\underline{z}_i - \underline{x}_i}{\underline{z} - \underline{x}_i}.$$

- ▶ Intuitively, $\bar{v}_{\pi,i}$ (resp. $\underline{v}_{\pi,i}$) is an upper (resp. lower) bound of $v_{\pi(i)}$ under π .
- ▶ To simplify the notation, we write $\bar{v}_i = \bar{v}_{\pi,i}$ and $\underline{v}_i = \underline{v}_{\pi,i}$ when $\pi(i) = i$ for each i .

MAIN THEOREM

- **Polytope** = a convex hull of finite points.

Theorem 1

1. The set of all feasible search behaviors V^* is a **polytope**.
2. v with $v_1 \geq \dots \geq v_n$ is a **vertex** of $V \Leftrightarrow v = (\bar{v}_1, \dots, \bar{v}_i, \underline{v}_{i+1}, \dots, \underline{v}_n)$ for some i .
3. A **single** CDF profile $\bar{F}_1, \dots, \bar{F}_n$ induces all feasible search behaviors, in which
 - Pandora is **indifferent** among all search orders, and
 - Pandora's expected payoff is **minimized** among all feasible CDF profiles.

FACES

- $X \subset V$ is a **face** of V if $X = \operatorname{argmax}_{v \in V} r \cdot v$ for some $r \in \mathbb{R}^n \setminus \{0\}$.

Proposition 1

Take any $r \in \mathbb{R}^n \setminus \{0\}$.

Let Π_r be the set of orders π such that

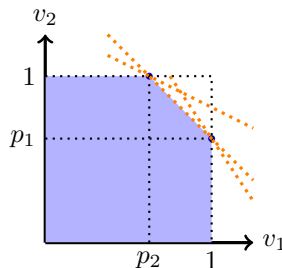
$$\begin{aligned} r_{\pi(1)} \cdot \frac{\bar{x}_{\pi(1)} - \underline{z}}{c_{\pi(1)}} &\geq \cdots \geq r_{\pi(i)} \cdot \frac{\bar{x}_{\pi(i)} - \underline{z}}{c_{\pi(i)}} \\ &\geq 0 \\ &> r_{\pi(i+1)} \cdot \frac{\underline{z} - \underline{x}_{\pi(i+1)}}{\underline{z}_{\pi(i+1)} - \underline{x}_{\pi(i+1)}} \geq \cdots \geq r_{\pi(n)} \cdot \frac{\underline{z} - \underline{x}_{\pi(n)}}{\underline{z}_{\pi(n)} - \underline{x}_{\pi(n)}}. \end{aligned}$$

Then, a face $X = \operatorname{argmax}_{v \in V^*} r \cdot v$ is the convex hull of all points v such that,

for some $\pi \in \Pi_r$, we have $v_{\pi(j)} = \bar{v}_{\pi,j}$ for all $j \leq i$ and $v_{\pi(j)} = \underline{v}_{\pi,j}$ for all $j > i$.

PROOF SKETCH I: UPPER BOUND

- ▶ We solved information design problems with linear objectives $\sum_i r_i v_i$ for any $r_i \geq 0$.
- ▶ This provides an **upper bound** for the set of feasible search behaviors.



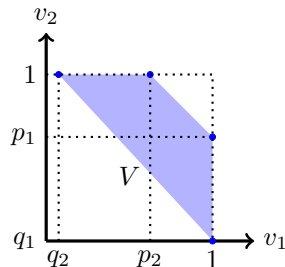
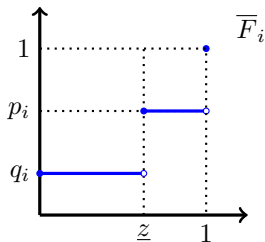
where

$$p_i = 1 - \frac{c_i}{\bar{x}_i - \underline{z}} = \max_{F_i: z_i \geq \underline{z}} F_i(\underline{z}),$$

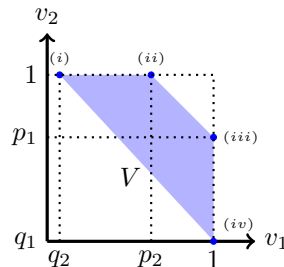
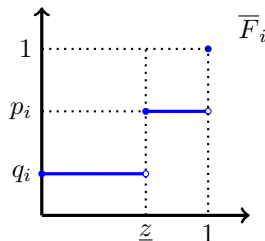
- ▶ Intuitively, we show that making z_i closer to $\underline{z}$ increases the search length.
- ▶ Trade-off: Increasing $z_i < \underline{z}$
 - ▶ raises the probability that box i is opened. (**attraction**)
 - ▶ makes box i more attractive; Pandora is more likely to terminate the search (**persuasion**)
- ▶ Challenge: v_i is discontinuous in CDF profiles as it also depends on the search order.
- ▶ A naive approach yields an ambiguous relation between these effects.
- ▶ However, we show that, under the order $\pi \in \Pi_r$, the first effect is dominant.

PROOF SKETCH III: CONSTRUCTION

- ▶ We construct $\overline{F}$ which **implements all vertexes** of V .
- ▶ Idea: Ensures that Pandora is indifferent among as many strategies as possible.
 - ▶ Reservation value = $\underline{z}$: **All search orders are optimal.**
 - ▶ A positive (maximum) mass on $\underline{z}$: **Indifferent btw Stopping and Continuing search.**

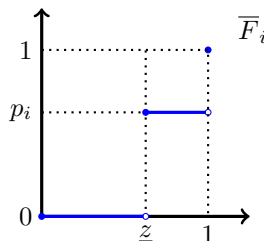


- ▶ Check that $\overline{F}$ implements all vertexes: hence all points in V .
- ▶ Under $\overline{F}_i$, there are three possible realizations: $y_i = 0, \underline{z}$, or 1.
 - ▶ (i) (resp. (iv)): Open box 2 (resp. 1) first and stop if y_2 (resp. y_1) is either 0 or $\underline{z}$.
 - ▶ (ii) (resp. (iii)): Open box 2 (resp. 1) first and stop if y_2 (resp. y_1) is 1.



PROOF SKETCH IV: WELFARE

- ▶ Pandora's expected payoff is **minimized** among all information structures.
- ▶ Consider a strategy that opens box i with $\underline{z}_i = \underline{z}$ and stops immediately.
- ▶ Feasible under any CDFs F , **guaranteeing** payoff $\mu_i - c_i = \underline{z}_i$.
- ▶ However, it turns out that it is a **Pandora's optimal strategy** under $\overline{F}$.



IMPLICATION I: ROBUST PREDICTION

- ▶ A metaphorical interpretation of the feasible set is about the **robust prediction**.
- ▶ In job search setting, potential workers sequentially inspect job opportunities.
- ▶ Requirement information may not perfectly reveal match quality.
- ▶ Our results: an information-free prediction for the range of possible search durations.
- ▶ This derives **unemployment periods**, which depend on the underlying information.

IMPLICATION II: INFORMATION DESIGN

- ▶ A literal interpretation is about the **optimal information design** on search behavior.
 - ▶ **Designer** chooses F with objective function $g : \mathbb{R}^n \rightarrow \mathbb{R}$ over search behaviors.
- ▶ Amazon and Yelp design information revealed within each product page.
 - ▶ e.g., page layouts, sellers' descriptions, and customer reviews to highlight.
- ▶ Consumers learn about product quality through these designs
 - ▶ However, they cannot fully infer quality until after they purchase and consume.
- ▶ Consumers' browsing behavior affects profits through per-click advertising fees.

Proposition 2

1. $\bar{F}$ is optimal for any upper continuous objective g over search behaviors, which **minimizes** Pandora's expected payoff among all feasible CDF profile.
2. Moreover, if $g(v) = r \cdot v$ for some $r \in \mathbb{R}^n \setminus \{0\}$, then
 - (i) it is optimal to let Pandora pen boxes according order in Π_r , and
 - (ii) if $\bar{z} > \underline{z}$, an optimal search behavior is fully implementable.

- Our results imply that information design **adversary affects consumer welfare**.
 - An extension shows that this persist even if designer also cares eventual choice.
- Designer-optimal search order is descending in $r_i \frac{\bar{x}_i - \underline{z}}{c_i}$ if $r_i \geq 0$ for all i .

IMPLICATION III: PARTIAL IDENTIFICATION

- ▶ Another one is about the **partial identification** with **weak informational assumption**.
- ▶ Suppose that analysts have access to data on search behavior v .
- ▶ They wish to estimate the primitives such as **search costs** and **preferences**.
- ▶ However, they do not know to what extent the agent learns from each inspection.
- ▶ Assumptions about information structure affects estimations and predictions.
- ▶ Our closed form characterization of the feasible set maps directly to the **identified set**.

- Let $\Gamma \subset \mathbb{R}^{4n}$ be the set of all parameters $(\bar{x}_i, \underline{x}_i, \lambda_i, c_i)$.
- **Identified set** is $I(v; \Gamma) = \{\gamma \in \Gamma \mid v \in V^*(\gamma)\}$, where $V^*(\gamma)$ is the feasible set under Γ .

Proposition 3

Take any search behavior v such that $v_1 \geq v_2 \geq \dots \geq v_n \geq 0$.

Then, the identified set for the **symmetric parameter space** Γ^* is

$$I(v; \Gamma^*) = \left\{ \gamma \in \Gamma^* \left| \sum_{i=1}^n v_i \geq 1 \text{ and } \sum_{j \leq k} v_j \leq \sum_{j \leq k} p(\gamma)^{j-1} \text{ for all } k \right. \right\},$$

where for each $\gamma = (\bar{x}, \underline{x}, \lambda, c)$, we set $p(\gamma) = 1 - c/(\bar{x} - \underline{x})$.

- Marginal of $I(v; \Gamma^*)$ is an interval $[0, \bar{c}]$ because $p(\gamma)$ is decreasing in c .

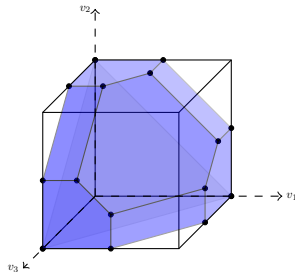
COMPARATIVE STATICS

- ▶ In any interpretations, how the feasible set varies with the primitives matters.
- ▶ By the closed-form expressions, we can trace how each vertex responds to changes.

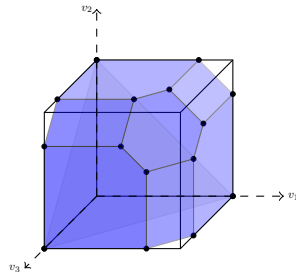
Proposition 4

1. If $\underline{z}_i = \underline{z}$, V^* **shrinks** when λ_i increases.
2. If $\underline{z}_i < \underline{z}$, then V^* **expands** when λ_i increases.
3. In the symmetric case where $\lambda_i = \lambda$, $\bar{x}_i = \bar{x}$, $\underline{x}_i = \underline{x}$, and $c_i = c$ for all i ,
 V^* **shrinks** when λ increases or c increases.

- ▶ The effect of increasing λ_i depends on the relative ex-ante attractiveness of box i .
- ▶ The effect of increasing c_i is also ambiguous.
- ▶ However, it turns out that the set is monotone in the symmetric primitives.



$\lambda = 0.8$ and $c = 0.2$.



$\lambda = 0.7$ and $c = 0.1$.

- ▶ As λ_i decreases, Pandora is more likely to sample a low value and continue searching.
- ▶ Hence, the maximum search length $r \cdot v$ weighted by $r \geq 0$ increases.
- ▶ Under the symmetric primitives, $\min r \cdot v$ does not depend on the primitives.
- ▶ Therefore, the range of $r \cdot v$ becomes larger, which suggests that V^* expands.
- ▶ If the primitives are asymmetric, $\min r \cdot v$ depend on the primitives,
in which case V^* may neither shrink nor expand.

EXTENSIONS

FEASIBLE CHOICE BEHAVIOR

- ▶ Search behavior $v \in \mathbb{R}^n$ is not a full description of Pandora's action.
- ▶ **Choice behavior** $d \in \mathbb{R}^n$: d_i is the probability that each box i is **eventually selected**.
- ▶ Question: Can we characterize the set of all **feasible pair** $(v, d) \in \mathbb{R}^n \times \mathbb{R}^n$?

Theorem 2

Suppose that $\underline{z}_i$ is homogeneous across boxes.

1. Then, the set of all feasible pairs S^* is a polytope.
2. Moreover, a single CDF profile F^* rationalizes all feasible pairs,
which minimizes Pandora's expected payoff among all feasible CDF profile.

- ▶ All vertexes are derived in closed form & Implications are **identical** to Theorem 1.

Proposition 5

Suppose that $\underline{z}_i$ is homogeneous across boxes.

(v, d) with $v_1 \geq \dots \geq v_n$ is a **vertex** of S^* if and only if, for some i and $j \leq i$, for each k ,

$$v_k = \begin{cases} \bar{v}_k & \text{if } k \leq i, \\ 0 & \text{otherwise,} \end{cases} \text{ and } d_k = \begin{cases} \bar{v}_k - \bar{v}_{k+1} & \text{if } k \leq i \text{ and } k \neq j, \\ \bar{v}_k - \bar{v}_{k+1} + p_i \cdot \bar{v}_i & \text{if } k = j, \\ 0 & \text{otherwise.} \end{cases}$$

Moreover, any vertex is fully implementable.

- ▶ Each vertex consists of three elements:
 - ▶ (i) a **search order** π , (ii) a **threshold box** i , and (iii) a box j for the "**return demand**."
- ▶ F_k^* takes binary values $\{\underline{z}, h_k\}$, and Pandora uses the following strategy.
 - ▶ She opens boxes in the order π , with the highest possible probabilities to boxes up to box i .
 - ▶ Pandora stops search and selects box k whenever she observes its high realization h_k .
 - ▶ She selects box j when all realizations up to box i are low ($y_k = \underline{z}$ for all $k \leq i$).
 - ▶ Boxes after i are never opened and therefore never chosen.

PRICE COMPETITION

- ▶ In consumer search, each box may be a product which is associated with its seller.
 - ▶ Consider **sellers who post prices** $t_i \geq 0$ to maximize revenue.
 - ▶ A CDF and an equilibrium pricing induce an **equilibrium search behavior**.
 - ▶ Issue: no characterization of equilibrium pricing is known.
 - ▶ However, we can still characterize equilibrium search (and choice) behaviors.

Theorem 3

Suppose that $\underline{z}_i$ is homogeneous across boxes.

1. The set of all eq. search behaviors is **same to** the set of feasible search behaviors.
2. A **single** CDF profile rationalizes all equilibrium search behaviors,
under which the sellers obtain **zero surplus**.

GENERAL STATE SPACE

- ▶ Reward x_i may not take binary values, following general prior $F_i^0 \in \Delta([\underline{x}_i, \bar{x}_i])$.
- ▶ In this case, F_i is feasible $\Leftrightarrow F_i$ is a **mean-preserving contraction** of F_i^0 , i.e.,

$$\int_{\underline{x}_i}^{x_i} F_i(y_i) dy_i \leq \int_{\underline{x}_i}^{x_i} F_i^0(y_i) dy_i \text{ for all } x_i \text{ with equality at } x_i = 1.$$

Proposition

Suppose that $\underline{z}_i$ is homogeneous across boxes.

1. The set of all search (and choice) behaviors is a **polytope**.
2. A **single** CDF profile rationalizes all search (and choice) behaviors, which **minimizes** Pandora's expected payoff among all feasible info. structures.

CONCLUSION

CONCLUSION AND REMARKS

- ▶ Question: Which search behavior is rationalized in search markets?
 - ▶ Add an element of **incomplete information** in Weitzman's (1979) model.
- ▶ Results: Deriving the set of all feasible search behaviors.
 - ▶ A **welfare-minimizing** CDF profile rationalizes any point in the feasible set.
 - ▶ Generalized to the set of all feasible **pairs** of search and choice behaviors.
 - ▶ We can also extend to include **price competitions** among sellers into the model.

THE END.

APPENDIX

FEASIBLE SET UNDER $n = 3$

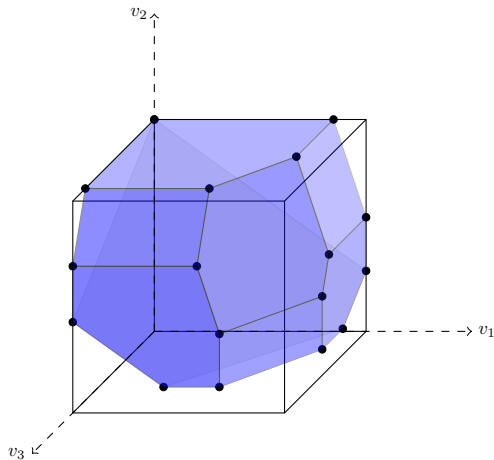


Figure: The set of feasible search behaviors V^* .

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